

Fire Incident Response Education, Guardianship and Preparedness

Together, we can bridge the gap in fire safety and preparedness for Alaska's most rural communities.

Project Overview

The Fire Incident Response Education, Guardianship, and Preparedness (FIRE GAP) program addresses critical gaps in fire prevention and response in rural Alaska. It empowers communities through tailored training, mentorship, and resource provision to manage both wildland and structural fires. Designed to build resilience in remote areas, the program emphasizes cultural relevance, community ownership, and a scalable approach to fire safety and preparedness.

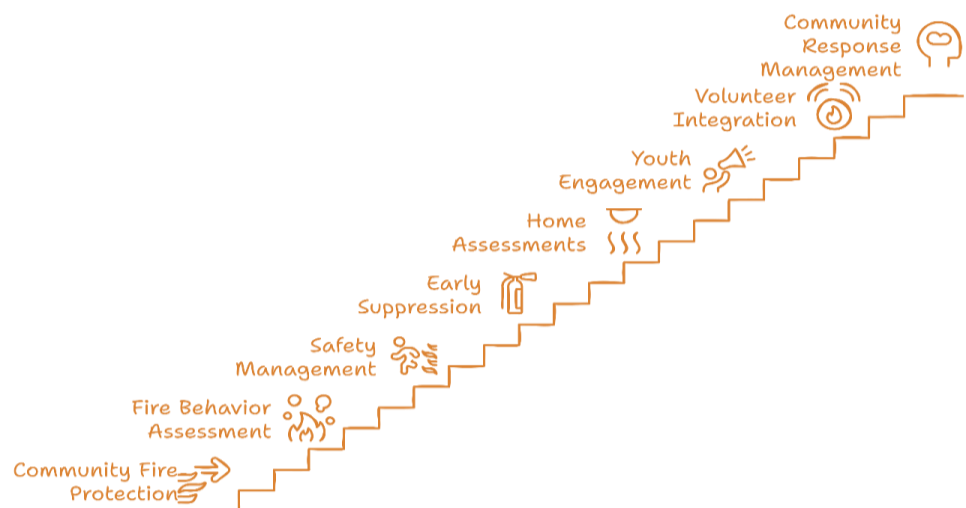


Impact Statement

The FIRE GAP program will:

- Enhance fire resilience in underserved communities by training local responders and educators.
- Mitigate fire risks through practical prevention strategies and equipment distribution.
- Foster community empowerment by integrating cultural practices and building leadership among youth and elders.
- Save lives and protect property by bridging critical response and knowledge gaps.

Steps to filling the FIRE GAP

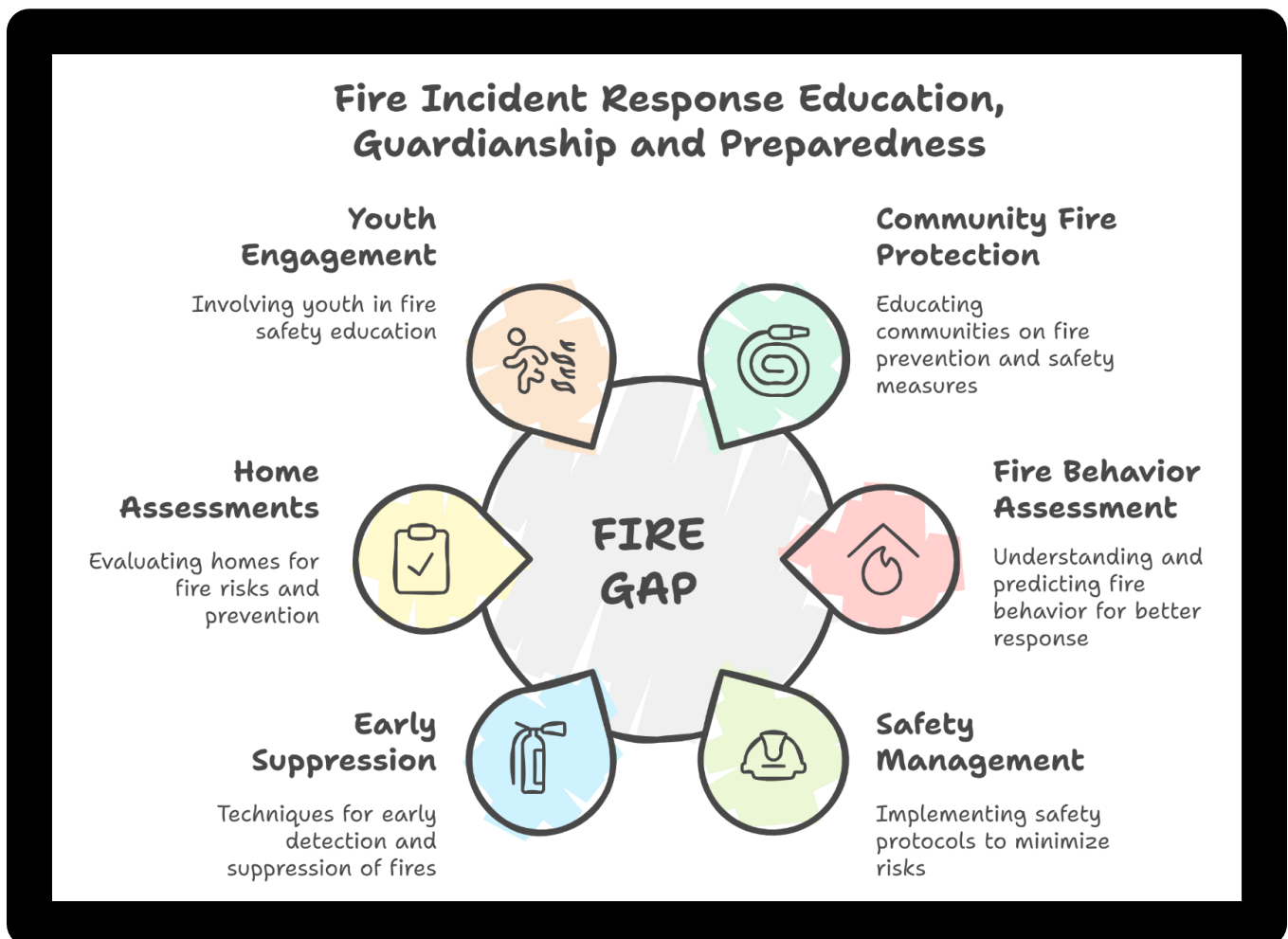


Introduction

Rural Alaska communities and Tribes face significant challenges in addressing disasters, including structural and wildland fires. The unique geographical, cultural, and economic landscape of Alaska complicates fire management and response efforts, creating a pressing need for enhanced capacity and resources. This proposal aims to explain the complex issues these communities face and provide potential funders with a comprehensive understanding of the current situation and new opportunities.

Alaska Venture Fund (AVF) is in a unique position to meet communities where they are to build response capabilities appropriate for the varying social, economic and political situations that exist in off-road, rural Alaska. AVF's wildfire team has extensive wildland fire management, rural fire prevention training, public education and community outreach experience. With deep connections across Alaska, we believe our community-based collaborative approach will better prepare and prevent disasters by increasing individual preparedness and mentoring those in communities looking to become more organized, pursue grant opportunities and gain employment as wildland firefighters.

A Fire Incident Response Education, Guardianship and Preparedness (FIRE GAP) pilot project could immediately begin to address those gaps related to fire prevention and response needs within these communities.



Challenge: Geographic and Climatic Context

Alaska's vast and remote landscape presents unique challenges for fire management. The state's rural and Tribal areas are often accessible only by air or water, which limits the rapid deployment of firefighting resources and the accessibility of mutual aid. The impacts of climate change exacerbate these challenges, with increasing temperatures, longer fire seasons, more lightning and changing precipitation patterns leading to more frequent and severe wildfires.



Red dots indicate Rural Alaskan communities that are unable to meet the minimum national standard of firefighting, NFPA 1010. These communities may lack human, financial, equipment and/or training resources. (State of Alaska, Office of Rural Fire Protection 2020)

Challenge: Fire Preparedness & Response

Wildfires

- **Increased Frequency and Severity:** Climate change has impacted Alaska's tundra and boreal forests, leading to increases in the frequency and severity of wildfires. This trend is expected to continue, putting more communities at risk. Alaska could see an increase in annual acres burned by as much as 150% by the year 2050. (Wildfire Strategy Report, 2022)
- **Fire Sources:** Alaska has seen an increase in lightning activity and more lightning caused fires with an increase in average acres burned. Human caused fires have also increased, often resulting in more annual starts than lightning.

***Threats to communities include
wildland to structure, structure to wildland and structure
to structure fires.***

- **Evacuation and Emergency Response:** In most rural communities, evacuation routes are often limited, complicating emergency response efforts. Additionally, most communities lack response resources and Community Wildfire Protection Plans (CWPP) for an effective defense and response. Evacuation and/or sheltering in place options are very limited. (Wildfire Strategy Report, 2022)



Structural Fires

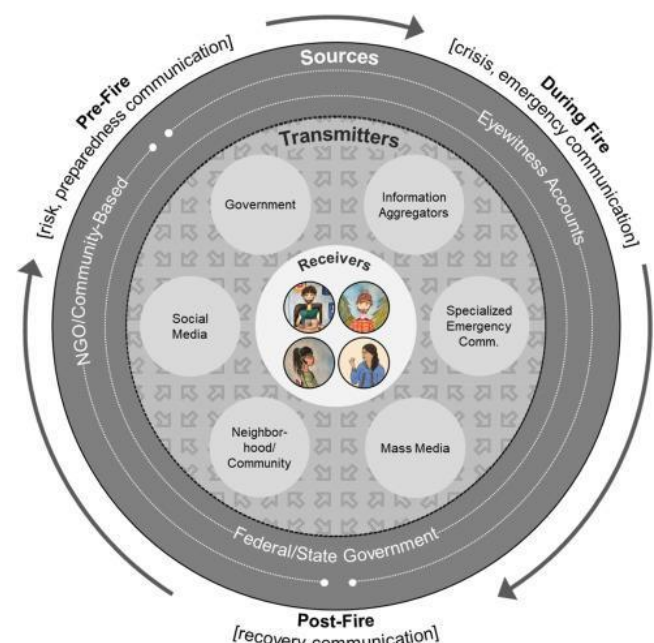
- **Training and Personnel:** Due to low populations and lack of adequate funding in most rural communities, there is an inability to meet the national minimum standard of training firefighting personnel. Untrained community responders often fill the gap, but they lack the necessary experience and training to respond effectively to structural fires and handle life threatening situations. (Herrmann, 2024)
- **Limited Infrastructure:** About 60% of Alaskan communities (170 out of 285) do not have a state-registered fire department (FD), limiting their opportunities for funding to obtain appropriate infrastructure. Of the 115 registered FDs, 65 cannot meet the national minimum standard, leaving 50 registered FDs that can. Regardless of registration status, nearly all 235 rural communities lack basic firefighting infrastructure, including fire stations, equipment, and reliable communication systems essential to meet the national minimum standard. (Fire Department Registration Status, 2024)
- **Accessibility:** In some cases, the limited infrastructure makes it difficult to quickly access water for firefighting. This often results in significant property damage and loss of life during emergencies. Remote communities often lack ground transportation routes, preventing mutual aid agreements and support from neighboring areas during an emergency. Poor road access and the absence of adequate water supply systems further hinder firefighting efforts. (Evans et al., 2007)

Meets Min. Standard?	Fatalities (2004-2023)	Total Population in Communities with Fire Deaths	Per Capital Overall (Per 1000 People)	Relative to Standard
Yes	168	398,325	0.42	-
Some	87	82,778	1.05	2.5 times
No	104	28,497	3.65	8.7 times

- **Disproportionate Fire Fatalities:** Based on a community's ability to sustain a fire department to the national minimum standard, those that are unable have endured **8.7 times more fire fatalities in the last 20 years** than those that meet, or sometimes meet, the minimum standard. (Alaska Civilian Fire Fatality Statistics, 1961 to Present)

Community Fire Education

- **Meeting Community Needs:** Fire safety education, including public health information is crucial in rural areas where access to emergency services is extremely limited. Communities are often unsure who to contact for specific fire training, prevention and information needs.
- **Public Information Access:** In rural Alaska, information channel confusion can be complicated by wildland and urban agency boundaries and overlap. Three agencies are responsible for wildland fire management in Alaska: the Bureau of Land Management Alaska Fire Service, the Alaska Division of Forestry and the U.S. Forest Service. Structural fire



management authority and fiscal responsibility belong to local entities, such as municipalities, boroughs, Tribes or non-profits, leaving little to no statewide public information dissemination.

- **Fire Education:** Wildland fire agencies and partners develop fire education and prevention through the Alaska Wildland Fire Coordinating Group, but its outreach in rural areas is limited. The Alaska State Fire Marshal's Office strives to ensure that fire safety messaging is consistent across registered fire departments. However, a significant number of communities, nearly 83% of which are rural, lack the resources and expertise needed to deliver effective fire safety education, contributing to a disproportionate rate of fire fatalities in these areas.
- **Trusted Messengers:** Fire communications for prevention, education and response need to reflect diverse audiences and be tailored for communities. (Garbis Z. et al, 2023)

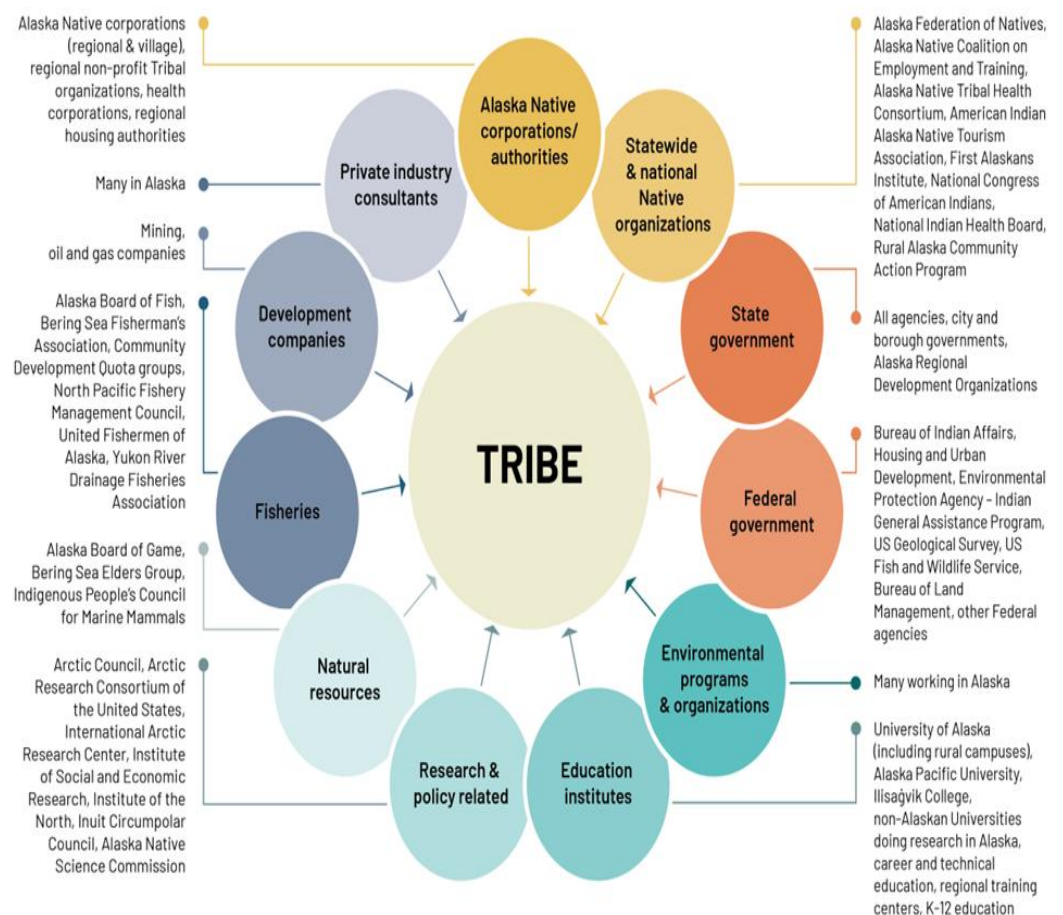
Fire protection starts at the home.

Challenge: Community Capacity Issues

Human Capacity

- **Limited Workforce:** The small population size in rural communities means that there are few people available to take on firefighting roles. Those who do are often stretched thin, managing multiple responsibilities beyond firefighting. (Herrmann, 2024)
- **Training Gaps:** There is a lack of accessible training programs tailored to the unique needs of Alaska Native communities and the specific environmental challenges they face (Wildfire Strategy Report,

A day in the life of an Alaskan Tribe



This illustration provides a snapshot of some of the external entities that Tribes may deal with on a daily basis. Groups listed depend on the specific Tribe and are subject to change. Last updated June 2024.

Credit: Herrmann, A. (2020). A day in the life of an Alaskan Tribe (graphic). Alaska Center for Climate Assessment and Policy. uaf-accap.org/projects/tribal-landscape

2022). Knowledge transfer between generations and the incorporation of traditional fire management practices are often limited (Evans et al., 2007). This gap is particularly pronounced in remote areas where travel to training centers is economically and logistically challenging.

Financial Capacity Constraints

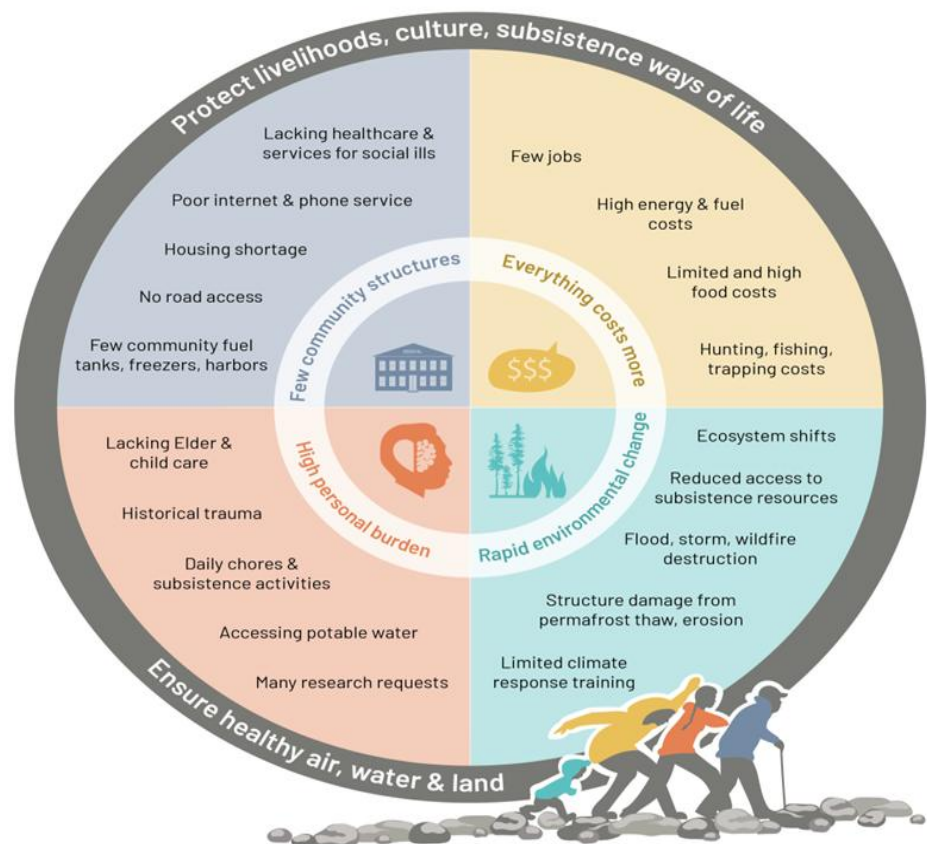
- **Insufficient Funding:** Many rural communities lack the financial resources to invest in necessary firefighting infrastructure and training. Federal and state funding opportunities exist but are often difficult to access due to complex application processes and stringent requirements. For example, Alaska Department of Forestry offers \$5,000 Volunteer Fire Assistance grants from U.S. Forest Service funds to better equip rural Alaska communities for wildland firefighting, but only registered fire departments qualify. Securing funding for firefighting equipment and training is a significant challenge. Grant writing and management require expertise and time that local leaders may not have. (Vulnerability and Adaptation Report, 2020)
- **Grant Management:** Even when funds are available, managing multiple grants and complying with reporting requirements can overwhelm the limited administrative capacities of rural communities and Tribes. (Herrmann, 2024)

Cultural and Social Challenges

- **Cultural Sensitivity:** Fire management practices must be culturally appropriate and inclusive of Indigenous knowledge and practices. Traditional ecological knowledge can play a crucial role in developing effective fire management strategies but is often underutilized. (Wildfire Strategy Report, 2022)

- **Community Engagement:** Engaging community members in fire preparedness and response activities is essential but challenging due to the daily stressors and responsibilities they face, including subsistence activities and limited access to basic services. (Herrmann, 2024)

Internal stressors Tribal members face



This graphic portrays some of the stressors Tribal or community members deal with in their personal lives, often on a daily basis. Examples listed depend on the specific Tribe, community or individual and are subject to change. Last updated June 2024. Credit: Herrmann, A. (2024). Internal stressors Tribal members face (graphic). Alaska Center for Climate Assessment and Policy, uaf-accap.org/projects/tribal-landscape

Opportunity: Fire Incident Response Education, Guardianship and Preparedness (FIRE GAP)

Alaska Venture Fund proposes offering an in-community training course (gap training) focused on rural Alaska community fire resiliency. This training will focus on community members preventing and managing structural and wildland fires, including community education and building a scalable training program. This program can include mentoring individual community members to pursue additional emergency response training certifications and guidance on how to acquire the necessary firefighting equipment for their communities. FIRE GAP will bridge the knowledge and skill gap from no response capabilities to understanding what is possible.

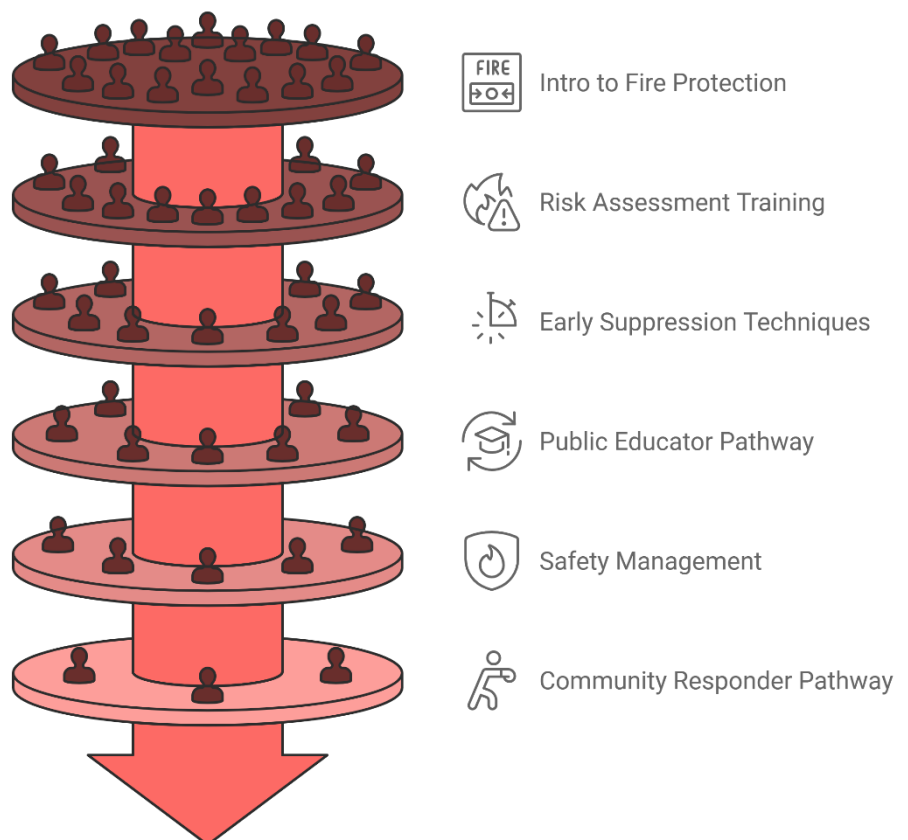
The FIRE GAP curriculum was developed using the American Council on Education (ACE) guide to be considered for credit for prior learning at higher education institutions. Course content references state and federal standards and industry best practices. Course content will be reviewed after every delivery to ensure that it remains relevant and adaptable to the community's needs.

The FIRE GAP aims to enhance human capacity by implementing tailored fire gap training programs in Alaska's rural and Tribal communities. Technical instructor teams, consisting of experts in wildland, structural, and public education will be trained to work in remote areas through a Train-the-Trainer course. This training will include comprehensive skill development and cultural competence to foster respectful and effective collaboration with Indigenous communities.

Alaska Venture Fund will engage with communities through targeted outreach and awareness campaigns, improving community response to local training. These technical teams will advocate for the use of educational resources and toolkits for fire risk management, as well as offer certification courses to attract and retain local volunteers and strengthen the statewide workforce. The program will empower rural communities in fire management planning and decision-making processes, incorporating traditional ecological knowledge to develop culturally relevant and effective strategies.



FIRE GAP Funnel

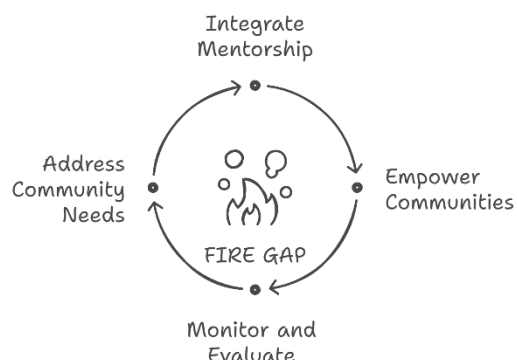


Components of this training will involve young children and youth to maximize a holistic approach to community engagement and foster potential future workforce opportunities.

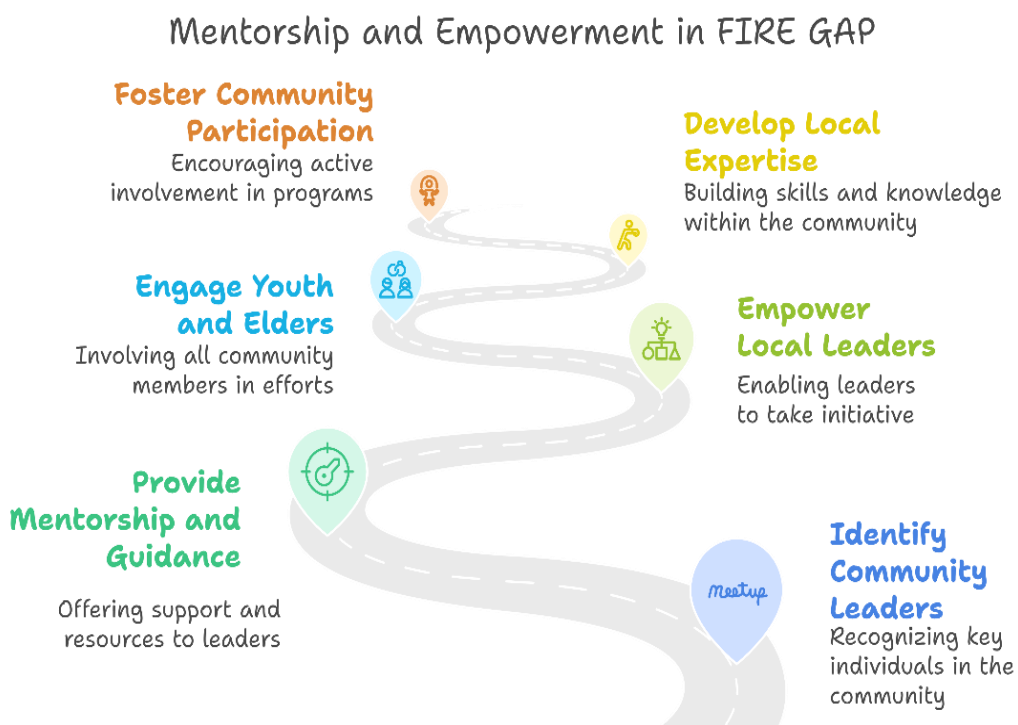
To enhance resources, appropriate fire prevention equipment will be provided based on the community's current capacity. This may include water extinguishers, backpack fire pumps, home fire blankets, battery-powered weed blowers, weed trimmers, chainsaws, and hedge trimmers. In collaboration with the American Red Cross, home assessments and training courses will be offered, including residential smoke alarm installations, as well as certified First Aid and CPR training. Additionally, in partnership with ANTHC's Healthy Homes project, installations of Purple Air Monitoring systems will be provided.

Program Sustainability

The sustainability of the FIRE GAP program depends on an interdisciplinary approach that ensures long-term viability, community ownership, and continuous impact. By integrating mentorship, community empowerment, and a strong focus on monitoring and evaluation, we aim to create a self-sustaining program that addresses the evolving needs of high-risk communities across Alaska.



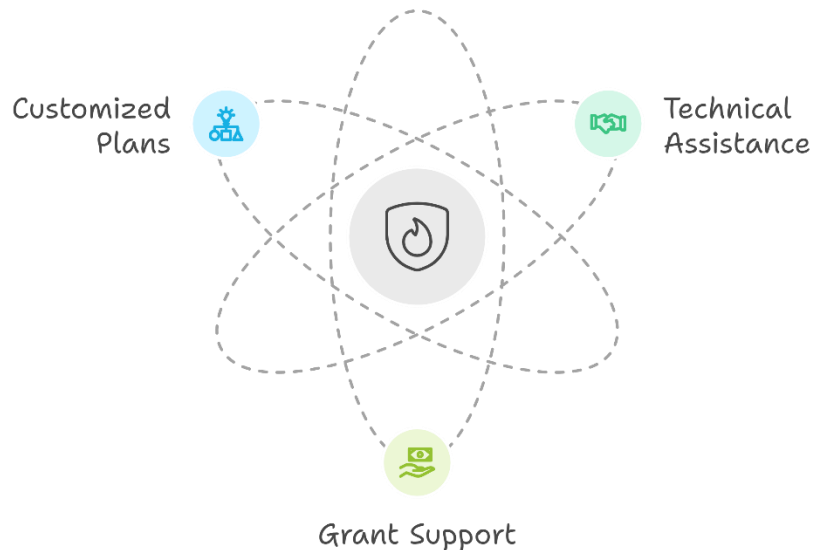
- **Mentorship:** To foster sustainability, the FIRE GAP will prioritize the empowerment of local community leaders, youth, Elders, and other members who play pivotal roles in fire adaptation efforts. By creating opportunities for these individuals to lead and direct initiatives, we will provide ongoing mentorship, guidance, and resources tailored to the unique challenges of rural Alaska. This approach ensures that communities are not only beneficiaries of the program but active participants in its development and evolution, building a foundation of local expertise and leadership.



- **Monitoring, Evaluation, and Community Engagement:** A robust framework for monitoring and evaluating the effectiveness of the fire adaptation strategies learned through the FIRE GAP will be established. This framework will include mechanisms for soliciting feedback from community

members, assessing outcomes, and making necessary adjustments to maintain relevance and effectiveness. Through targeted outreach and awareness campaigns, we will actively engage with communities, introducing additional educational resources and toolkits for fire risk management. Community meetings, workshops, and webinars will provide platforms for continuous improvement and adaptation of the strategies to local needs.

- **Technical Assistance and Grant Support:** AVF will offer ongoing technical assistance and support to ensure the successful implementation of plans created in the FIRE GAP. By providing communities with guidance on adapting to their local contexts and conducting site-specific risk assessments, communities will be empowered to develop customized structural and wildfire action plans. Additionally, we will conduct thorough research to identify and assist communities to secure grant funding opportunities that align with the strategic objectives of community-based wildfire protection planning, ensuring a steady flow of resources to sustain their identified plans.



- **Partnership Building and Knowledge Sharing:** The program's sustainability will be further enhanced through strategic partnerships and collaborations between Tribes, local governments, and non-profit organizations. Sharing lessons learned, best practices, and success stories will foster a culture of knowledge sharing and collaboration, encouraging replication of successful strategies across different communities and regions.



Summary

Addressing fire management challenges encountered by rural Alaska communities and Tribes necessitates a comprehensive approach that includes assessing the current capabilities of each community, enhancing or providing essential suppression equipment, and imparting critical knowledge related to structural and wildland fire management.

Our strategy is designed to support communities by equipping them with foundational knowledge and tools to safeguard their residences, subsistence areas and improve community resiliency in the face of fire. By securing funding for the Alaska Community Fire Protection pilot project, we can immediately and significantly enhance the safety and resilience of rural Alaska communities, with the potential to scale and expand training and access to rural Alaska communities in additional regions.

Alaska Venture Fund

Strengthening Wildfire Resilience in Alaska

Coalition Building. Collaborative Change.

Alaska Venture Fund is a philanthropic partner and social-change incubator building a more sustainable future for Alaska and beyond. Our goal is to create solutions with broad-based impact, building powerful coalitions across communities and cultures.



Our Alaska Wildfire Resilience Program

We are working with community and tribal leaders to improve resilience and safety in the face of increasing risk of severe and catastrophic wildfire in Alaska. The program is leveraging significant public and private funding for pre-fire and early fire preparedness and intervention, and is building innovative and bold relationships with stakeholders across Alaska's diverse landscapes and throughout the wildfire sector.

wildfire@alaskaventure.org | alaskaventure.org |  Alaska Venture Fund

For more information, please contact:

Jeff Ennenga

Wildfire Resilience Program
Director

971-317-1752

jeff@alaskaventure.org

Lisa Shield

Community Wildfire Resilience
and Workforce Coordinator

907-775-5485

lisa@alaskaventure.org

Kenni Psenak

Wildfire Outreach and
Communications Manager

907-982-4304

kenni@alaskaventure.org

References

1. *Alaska Civilian Fire Fatality Statistics, (1961 to Present)* State of Alaska, Department of Public Safety, Fire and Life Safety.
2. *Fire Department Registration Status, (2024 Webdata)* State of Alaska, Department of Public Safety, Fire and Life Safety.
3. Herrmann, A. (2024). [Landscape of Tribal Communities: Context for Working in Rural Alaska](#). Alaska Center for Climate Assessment and Policy, UAF International Arctic Research Center.
4. *Wildfire Strategy Report, (2022). The Role of First Nations Guardians in Wildfire Response & Management: A Proposed National Strategy*. Centre for Northern Conservation, International Boreal Conservation Campaign, McCall MacBain Foundation.
5. Evans, A., et al. (2007). *Community Capacity and Wildfire Preparedness: A Review of Literature and Case Studies*.
6. Rutherford, J. (2019). *Adapting Wildland Fire Management to 21st Century Conditions*.
7. *Vulnerability and Adaptation Report, (2020). Vulnerability and Adaptation to Climate-Related Fire Impacts in Rural and Urban Interior Alaska*.
8. *Barriers Discussion Paper, (2023). Barriers to Effective Fire Management in Indigenous Communities*.
9. Alaska Fire Science Consortium, (2020). [Alaska's Changing Wildfire Environment](#)
10. Garbis, Z., et al. (2023). [Taming the wildfire infosphere in Interior Alaska: Tailoring risk and crisis communications to specific audiences](#), [International Journal of Disaster Risk Reduction](#)